B4.5.17

$$y'' - 2y' - 3y = 3t^2 - 5$$

Characteristic equation of complementary equation is (again check/find this first in case there is overlap):

$$r^2 - 2r - 3 = 0$$
$$(r - 3)(r + 1) = 0$$
$$r = -1, 3$$

Thus

$$y_c = c_1 e^{-t} + c_2 e^{3t}$$

Definitely no overlap here so try the usual

$$y_p = At^2 + Bt + C$$

From which

$$y_p' = 2At + B$$

$$y_p'' = 2A$$

Plugging these all in the original equation gives:

$$2A - 2(2At + B) - 3(At^2 + Bt + C) = 3t^2 - 5$$

Equate coefficients to yield:

$$t^2: \ -3A = 3$$
$$t: \ -4A - 3B = 0$$
$$\text{constant}: \ 2A - 2B - 3C = -5$$

From first, $A = -1$, then from second $B = \frac{4}{3}$, and then

$C = \frac{1}{9}$

So our general solution is

$$y(t) = c_1 e^{-t} + c_2 e^{3t} - t^2 + \frac{4}{3} t + \frac{1}{9}$$

Why does this always work for polynomial? Well this doesn't tell full story, but those equations solve for each constant, one at a time, sort of a triangular system with no zero diagonal entries, invertible!

B4.5.21

$$y''(t) + 2y'(t) + 2y(t) = e^{-t} \cos t$$

Characteristic equation of complementary equation is

$$r^2 + 2r + 2 = 0$$

---> $$(r + 1)^2 + 1 = 0$$

---> $$r = 1 \pm i$$

Thus

$$y_c = c_1 e^{-t} \cos t + c_2 e^{-t} \sin t$$

Overlaps (root is not repeated so single overlap) meaning we try usual guess multplied by t:

$$y_p = Ate^{-t} \cos t + Bte^{-t} \sin t$$

Still will be messy here though with each term having three things multiplied, second derivative will take a lot of calculation!

B4.6 Variation of Parameters

Still want to solve:

$$ay'' + by' + cy = f(t)$$

But what if f is not of type where undetermined coefficients works, e.g. $f(t) = \tan t$ or $f(t) = \ln t$.

That's what variation of parameters is for.

So in a sense this is better than undetermined coefficients as it works in all cases where undetermined coefficients works plus some others.

Might be viewed as a universal approach for solving second order linear differential equations with constant coefficients, but realistically it's very delicate.

That is, while you can try this method for any $f(t)$, the way it tries to resolve (as we will see) is via doing some integrals. But very often those integrals will be impossible or at the least very hard is the issue.

If you try to make up a problem on your own, one where undetermined coefficients doesn't work, that's usually what will happen. These problems really have to be designed to work, again very delicate.

But interestingly, this method can even work sometimes in nonconstant coefficients which is something we haven't even considered yet.

Shall see that later on, but first we establish in the more familiar setting.

For $ay'' + by' + cy = f(t)$, suppose

$$y_c = c_1 y_1(t) + c_2 y_2(t)$$

is the general solution to the complementary equation. That is, y_1, y_2 are the two linearly independent solutions. Remember $f(t)$ has no effect on that part of it.

For the particular solution try now:

$$y_p = u_1(t)y_1(t) + u_2(t)y_2(t)$$

That is, similar form as y_c just now the coefficients of y_1, y_2 are functions not constants.

Why is this logical? Well y_1, y_2 being solutions to

$$ay'' + by' + cy = 0$$

we suspect that when we plug in the proposed solution that many things will cancel, and the equation might be reasonable to solve from there. But ok a lot to that.

Note first that

$$y_p' = u_1' y_1 + u_2' y_2 + u_1 y_1' + u_2 y_2'$$

Important to understand that y_1, y_2 are fixed here, even if unspecified, u_1, u_2 are the unknowns, those are what we are trying to solve for.

So say we now proceed naturally by computing y_p''

Quite clearly that will be a big mess. When we plug that all back into the desired equation, sure some stuff may cancel, but it's definitely not going to be simple, likely it'll be worse! And you will now have two unknowns u_1, u_2.

So we simplify this. If you learned variation of parameters in calculus, you likely saw this trick, but it probably wasn't very motivated. The motivation really lies in linear algebra ideas which now we are of course

more familiar with.

We assume that:

$$u_1' y_1 + u_2' y_2 = 0$$

Why do this? Well makes two or four terms in above y_p' equal to 0 so both it and y_p'' are easier is a natural thing.

But the big thing is when you compute y_p'' there no longer will be u_1'' or u_2'' in the expression.

That is, in the overall differential equation we wish to solve, the two unknowns, u_1, u_2 no longer have their second derivatives in the equation so the equation is first order not second order, obviously more accessible to us!

So what is the downside? Since we set something to 0, somewhat arbitrarily it seems, there has to be a consequence to that.

Key is we will have to satisfy that equation too as well as the original, so we now have to solve a system of two equations rather than just one.

So would we rather solve a single second order differential equation or a system of two first order differential equations? Turns out the latter tends to be more accessible to us, as second derivatives really make things much harder on us. And system we are better now equipped to deal with via linear algebra ideas.

So under this assumption we have:

$$y_p'' = u_1' y_1' + u_2' y_2' + u_1 y_1'' + u_2 y_2''$$

Now plug everything into $ay'' + by' + cy = f(t)$ to get

$$a(u'_1 y'_1 + u'_2 y'_2 + \underline{u_1 y''_1} + \underline{u_2 y''_2}) + b(\underline{u_1 y'_1} + \underline{u_2 y'_2}) + c(\underline{u_1 y_1} + \underline{u_2 y_2}) = f(t)$$

Terms underlined in same color sum to 0 since y_1, y_2 are solutions to complementary equation. Hence why we tried this style of solution to begin with as we suspected it might simplify nicely. So this becomes

$$a(u'_1 y'_1 + u'_2 y'_2) = f(t)$$

Hence the overall system we must solve is

$$u'_1 y_1 + u'_2 y_2 = 0$$

$$u'_1 y'_1 + u'_2 y'_2 = \frac{f(t)}{a}$$

So solve this 2 x 2 system via elimination of variables, e.g. multiply the first equation by y'_1 and the second by y_1 then subtract them. Will yield:

$$u_1' = \frac{-fy_2}{a[y_1 y_2' - y_1' y_2]}$$

and

$$u_2' = \frac{fy_1}{a[y_1 y_2' - y_1' y_2]}$$

Can also look at this as:

$$\underbrace{\begin{bmatrix} y_1 & y_2 \\ y_1' & y_2' \end{bmatrix}}_{A} \begin{bmatrix} u_1' \\ u_2' \end{bmatrix} = \underbrace{\begin{bmatrix} 0 \\ \frac{f(t)}{a} \end{bmatrix}}_{b}$$

Can solve via inverse or Cramer's Rule (interesting use of that, definitely comes up where you don't expect it!):

$$u_1' = \frac{\det A_1(b)}{\det A} = \frac{\begin{vmatrix} 0 & y_2 \\ \frac{f(t)}{a} & y_2' \end{vmatrix}}{\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}}$$

$$u_2' = \frac{\det A_2(b)}{\det A} = \frac{\begin{vmatrix} y_1 & 0 \\ y_1' & \frac{f(t)}{a} \end{vmatrix}}{\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}}$$

which simplify to same formulas as earlier.

Remember $\begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}$ is the <u>Wronskian</u>

Hence you see one use of that. If you were solving a third order equation, might use the 3 x 3 Wronskian? Ok we can't really even think about doing that, second order is hard enough as is, but that quantity is more important than you might initially think.

You can use the formulas for u_1', u_2' directly, don't have to derive from scratch. But important to realize that these only solve for u_1', u_2', we still need to integrate to find both.

And that's the issue. Since y_1, y_2 are e^t, cos t, or $e^t \cos t$ style functions, and f(t) is often quite different (since we are assuming undetermined coefficients doesn't work), those integrals are often very hard or impossible which is why problems usually have to be specfically designed to make them work reasonably.

B4.6.1

$$y'' + 4y = \tan(2t)$$

Characteristic equation of complementary equation is:

$$r^2 + 4 = 0$$

---> $$r = \pm 2i$$

Thus

$$y_c = c_1 \cos(2t) + c_2 \sin(2t)$$

So we try

$$y_p = u_1(t) \cos(2t) + u_2(t) \sin(2t)$$

Via usual formula gives

$$u_1' = \frac{\begin{vmatrix} 0 & \sin(2t) \\ \tan(2t) & 2\cos(2t) \end{vmatrix}}{\begin{vmatrix} \cos(2t) & \sin(2t) \\ -2\sin(2t) & 2\cos(2t) \end{vmatrix}}$$

And since

$$2\cos^2(2t) + 2\sin^2(2t) = 2$$

this reduces to

$$u_1' = \frac{-\sin(2t) \cdot \tan(2t)}{2}$$

So therefore

$$u_1 = \frac{-1}{2} \int \sin(2t) \cdot \tan(2t) \; dt$$

$$= \frac{-1}{2} \int \frac{\sin^2(2t)}{\cos(2t)} \, dt$$

$$= \frac{-1}{2} \int \frac{1 - \cos^2(2t)}{\cos(2t)} \, dt$$

Via distributing fraction:

$$= \frac{-1}{2} \int [\sec(2t) - \cos(2t)] \, dt$$

Now both well known so we get:

$$u_1 = \frac{-1}{4} \left[\ln|\sec(2t) + \tan(2t)| - \sin(2t) \right]$$

Then

$$u_2' = \frac{\begin{vmatrix} \cos(2t) & 0 \\ -2\sin(2t) & \tan(2t) \end{vmatrix}}{2} = \frac{\sin(2t)}{2}$$

----> $$u_2 = \frac{-\cos(2t)}{4}$$

So because

$$y_p = u_1(t)\cos(2t) + u_2(t)\sin(2t)$$

we get

$$y_p = \frac{[\ln|\sec(2t) + \tan(2t)|]\cos(2t)}{-4}$$

Then general solution is

$$y(t) = y_c + y_p$$

that is:

$$y(t) = c_1 \cos(2t) + c_2 \sin(2t) + \frac{[\ln|\sec(2t) + \tan(2t)|]\cos(2t)}{-4}$$

So reasonably involved, integrals not trivial! Have to be able to do trig integrals here. At least terms on bottom of u_1', u_2' resolve well based on what y_1, y_2 are. But since f can be anything, top is unpredictable.

This problem is a clear indication of the delicacy though. The u_2 integral worked out nicely due to the cancellation between $\cos(2t)$ and $\tan(2t)$.

If original problem was solving equal to $\tan t$ or $\tan(3t)$, problem would be much harder! Again delicate, a slight change can make problem much harder or even impossible.

B4.6.7

$$y'' + 4y' + 4y = e^{-2t} \ln t$$

Characteristic equation of complementary equation is:

$$r^2 + 4r + 4 = 0$$

----> $$(r + 2)^2 = 0$$

----> $$r = -2, -2$$

Thus we have

$$y_c = c_1 e^{-2t} + c_2 te^{-2t}$$

So we try

$$y_c = u_1(t)\, e^{-2t} + u_2(t)\, te^{-2t}$$

Hence

$$u_1' = \frac{\begin{vmatrix} 0 & te^{-2t} \\ e^{-2t}\ln t & (-2t+1)e^{-2t} \end{vmatrix}}{\begin{vmatrix} e^{-2t} & te^{-2t} \\ -2e^{-2t} & (-2t+1)e^{-2t} \end{vmatrix}}$$

Denominator here is:

$$e^{-2t}(-2t+1)e^{-2t} + 2e^{-2t} \cdot te^{-2t} = e^{-4t}$$

Once again note that denominator simplifies fairly cleanly. So now

$$u_1' = \frac{-te^{-4t} \cdot \ln t}{e^{-4t}} = -t \cdot \ln t$$

Meaning

$$u_1 = -\int t \cdot \ln t \, dt = -\left[\frac{t^2}{2} \cdot \ln t - \int \frac{t^2}{2} \cdot \frac{1}{t} \, dt\right]$$

via by parts which then simplifies to

$$u_1 = \frac{-t^2}{2} \ln t - \frac{t^2}{4}$$

For the other:

$$u_1' = \frac{\begin{vmatrix} e^{-2t} & 0 \\ -2e^{-2t} & e^{-2t} \ln t \end{vmatrix}}{e^{-4t}} = \frac{e^{-4t} \ln t}{e^{-4t}}$$

Thus $u_2' = \ln t$ from which

$$u_2 = \int \underbrace{1} \cdot \underbrace{\ln t} \, dt = \left[t \cdot \ln t - \int t \cdot \frac{1}{t} \, dt \right]$$

$$= t \cdot \ln t - t$$

via by parts again. Hence

$$y_p = \left(\frac{-t^2}{2}\ln t - \frac{t^2}{4}\right)e^{-2t} + (t \cdot \ln t - t)te^{-2t}$$

$$= \left(\frac{t^2}{2}\ln t - \frac{3t^2}{4}\right)e^{-2t}$$

Thus

$$y(t) = y_c + y_p$$

gives general solution

$$y(t) = c_1 e^{-2t} + c_2 te^{-2t} + \left(\frac{t^2}{2}\ln t - \frac{3t^2}{4}\right)e^{-2t}$$

Here again very delicate as the e^{-2t} in solution we are trying to create cancels the exponential term in the Wronskian. If say we had e^{-t} or e^{-3t} instead problem probably becomes impossible.

Again gotta be on top of integration ideas, here we used by parts, last problem trig integrals, both ideas quite common in variation of parameters.

All the "advanced" integration ideas from Math 52 could come up in this topic, but definitely by parts and trig integrals are the biggest (they also both come up in the final chapter, B10, as well).

Note that when doing variation of parameters when finding u_1, u_2 via integration, you don't need to include the $+C$ that you usually have with indefinite integrals. Point is you need one "particular solution" so no need to try to "generalize" the particular solution that way.

If you did, that extra part, when you put this into the general solution would just mirror the homogeneous part of the solution anyway so would not change anything.

Although variation of parameters seems like a very restricted extension of undetermined coefficients since it only resolves reasonably in very specific situations where undetermined coefficients doesn't work, it actually can be used sometimes in situations where you would not initially expect it to help.

That's why it's more important than you might think at first. Will go over an example of that type next time.